

# Two sharp obstructions to the Boutet de Monvel conjecture: curvature, reversibility, and a flat-torus characterization

Candidate counterexamples for arXiv:2110.09389  
Run `fa_banach_001`, lane 10, model GPT5.6

22 July 2026

**Status.** Candidate counterexample likely valid, independently of the meaning assigned to “strictly convex.” The strongly convex multiplier  $P = (-\Delta)^{1/2} + cD_1$  on  $\mathbb{T}^n$  satisfies every printed hypothesis of Winterrose’s Conjecture 4.1, including everywhere positive curvature of its unit co-ball, but conclusion (4) fails. A second example,  $P = \partial_1^4 + \partial_2^4$ , shows that segment-strict convexity is also too weak for conclusions (1)–(2). For translation-invariant classical torus multipliers, all six conclusions hold exactly when the principal gauge has positive-definite fundamental Hessian and is reversible.

## 1 Source statement and the terminology audit

The target is Conjecture 4.1 of D. S. Winterrose, *Algebras of pseudo-differential operators acting on holomorphic Sobolev spaces*, arXiv:2110.09389, published in the *Journal of Functional Analysis* in 2023 [1]. It concerns an analytic, nonnegative, formally self-adjoint elliptic  $P \in \Psi_{\text{phg}}^d(M)$  with positive principal symbol  $p$ . It assumes that

$$\{\xi \in T_x^*M : p(x, \xi) \leq 1\} \text{ is strictly convex for every } x$$

and makes six assertions: the prescribed positive imaginary-time Hamilton flow is a real-analytic tube diffeomorphism; the tube has real-analytic boundary; it is strictly pseudoconvex; the kernel of  $e^{-\varepsilon P^{1/d}}$  extends holomorphically to it; its boundary value is a complex-phase PHG FIO of order  $-(n-1)/4$ ; and this FIO is the asserted holomorphic Sobolev homeomorphism. The complete source statement is shown in Figure 1.

A full source and version audit does not settle the meaning of “strictly convex.” The evidence is as follows.

- (1) All three arXiv versions use the displayed set-theoretic phrase. None defines it, mentions a Hessian or second fundamental form, or states a curvature lower bound. The only other substantive occurrence imposes the same condition on  $\{|a(x, \xi)| \leq 1\}$  when applying the conjecture.
- (2) Boutet de Monvel’s original two-page note likewise says that the *set*  $\{\sigma(P) \leq 1\}$  is “strictement convexe” [2]; see Figure 2. Under the usual convex-set convention this means absence of boundary segments.
- (3) The mathematical mechanism points the other way. The original note immediately claims a smooth strictly pseudoconvex tube and describes the general case as an easy perturbation of

**4. The Boutet de Monvel Conjecture.** (A theorem in a prototypical case.)  
 A theorem was announced by Boutet de Monvel in an old 1978 conference paper [2], but only the case  $\sqrt{-\Delta_g}$  in [2] was proved by Stenzel in [17] and Zelditch in [23]. Several possible approaches to proving it generally are discussed by Zelditch in [23]. To our knowledge, the general statement is not yet fully proven.

Let  $d > 0$ , and let  $P \in \Psi_{\text{phg}}^d(M)$  be analytic with classical principal symbol  $p$ . Suppose that  $P$  is formally self-adjoint, elliptic,  $p|_{T^*M \setminus 0} > 0$ , and that

$$\{\xi \in T_x^*M \mid p(x, \xi) \leq 1\} \text{ is strictly convex for all } x \in M.$$

In that case,  $P$  is semi-lower bounded, and we assume it has no negative eigenvalues. A real-analytic complete Hamiltonian flow  $\varphi_t : T^*M \setminus 0 \rightarrow T^*M \setminus 0$  arises from  $p^{\frac{1}{d}}$ . Given any  $t \in \mathbb{R}$  and  $(x, \xi) \in T^*M \setminus 0$ , it satisfies

$$\varphi_t(x, s\xi) = s \cdot \varphi_t(x, \xi) \text{ for all } s > 0,$$

and  $\varphi_t(x, \xi)$  extends holomorphically in  $t$  to  $\{t \in \mathbb{C} \mid |t| < \epsilon\}$  for  $\epsilon > 0$  small enough. The extension is then a map

$$\{t \in \mathbb{C} \mid |t| < \epsilon\} \rightarrow T_{1,0}^*M_{\mathbb{C}} \setminus 0 : t \mapsto \varphi_t(x, \xi),$$

where  $T_{1,0}^*M_{\mathbb{C}}$  is the holomorphic cotangent bundle of  $M_{\mathbb{C}}$ .

CONJECTURE 4.1 (Boutet de Monvel [2]. The statement here is a bit different).  
 Let  $P_{\epsilon}$  denote the Schwartz kernel of  $e^{-\epsilon P^{\frac{1}{d}}}$ , and put

$$\Phi_x(\xi) = (\pi_x \varphi_{\text{ip}(x, \xi)^{\frac{1}{d}}})(x, \xi) \text{ for any } (x, \xi) \in B_{\epsilon}^*M \setminus 0,$$

where  $\pi_x$  is the bundle projection, and

$$B_{\epsilon}^*M = \{(x, \xi) \in T^*M \mid p(x, \xi)^{\frac{1}{d}} < \epsilon\}.$$

There is a maximal  $\epsilon_0 > 0$  such that for any  $\epsilon \in (0, \epsilon_0)$  the following holds:

1. The extended flows combine into a real-analytic diffeomorphism

$$\Phi : B_{\epsilon}^*M \setminus 0 \rightarrow M_{\epsilon}^{\Phi} \setminus M \subset M_{\mathbb{C}} : (x, \xi) \mapsto \Phi_x(\xi).$$

2. The set  $M_{\epsilon}^{\Phi}$  is open in  $M_{\mathbb{C}}$  with (orientable)  $C^{\omega}$ -boundary  $\partial M_{\epsilon}^{\Phi}$ .
3. The set  $M_{\epsilon}^{\Phi}$  is strictly pseudo-convex.

(Point 3. implies that  $M_{\epsilon}^{\Phi}$  admits a Kähler metric  $h$ , and a global potential for it.)  
 There is a maximal  $\epsilon'_0 \in (0, \epsilon_0]$  such that for any  $\epsilon \in (0, \epsilon'_0)$  the following holds:

4. The map  $x \mapsto P_{\epsilon}(x, y)$  extends holomorphically to  $M_{\epsilon}^{\Phi}$  for each fixed  $y \in M$ .
5. The kernel  $P_{\epsilon}|_{\partial M_{\epsilon}^{\Phi} \times M}$  induces a complex-phase PHG FIO  $S_{\epsilon}$  of order  $-\frac{n-1}{4}$ .
6.  $S_{\epsilon}$  defines a homeomorphism  $S_{\epsilon} : H^s(M) \rightarrow \mathcal{O}^{s+\frac{n-1}{4}}(\partial M_{\epsilon}^{\Phi})$  for any  $s \in \mathbb{R}$ .

The first three points simply generalize the basic properties of the Grauert tubes. In the special case that  $P = -\Delta_g$ ,  $M_{\epsilon}^{\Phi}$  is exactly the Grauert tube  $M_{\epsilon}$ .

Figure 1: Conjecture 4.1 in arXiv:2110.09389v3, printed page 8.

the convolution case. Such a perturbation argument requires nondegenerate Legendre curvature and is false for a flat  $\ell^4$  axis.

- (4) Winterrose’s 2021 thesis gives a longer “Towards a Proof” discussion [3, Secs. 7.1–7.2]. Its differential-operator formulation omits a convexity hypothesis altogether, and the discussion concentrates on analytic FIO parametrices rather than defining the missing geometry. Thus it provides no terminological resolution.

The honest conclusion is that the printed statement is ambiguous: its words support segment-strictness, while its intended geometric mechanism likely needs positive curvature. The first counterexample below depends on that choice; the second does not.

## 2 Proof intuition

For a translation-invariant principal root  $q = p^{1/d}$ , put  $F = q^2/2$ . The conjectured positive imaginary-time flow is simply

$$\Phi(x, \xi) = x + i\nabla F(\xi).$$

Its Jacobian contains  $D^2F$ , so segment-strictness without positive curvature cannot give the claimed diffeomorphism. This produces the  $\ell^4$  counterexample.

There is a separate sign obstruction. If  $q$  is not reversible, the Legendre image of the positive imaginary flow is

$$D_\varepsilon^+ = \{v : q^*(v) < \varepsilon\},$$

whereas the Fourier factor  $e^{-\varepsilon q(k)} e^{ik(x+iv)}$  decays precisely on

$$D_\varepsilon^- = \{v : q^*(-v) < \varepsilon\} = -D_\varepsilon^+.$$

The two tubes agree exactly when  $q(\xi) = q(-\xi)$ . A Randers gauge  $q(\xi) = |\xi| + c\xi_1$  has an ellipsoidal, everywhere positively curved unit ball but is not reversible, so the kernel grows exponentially at points of the conjectured tube. This defeats the strong-curvature interpretation as well.

The same two observations are also sufficient. Positive  $D^2F$  gives the Legendre diffeomorphism and a nondegenerate complex phase; reversibility makes the flow tube equal the Fourier holomorphy tube. A lower-order classical term only multiplies the boundary FIO by an elliptic order-zero amplitude. This yields the exact flat-torus characterization below.

## 3 A positive-curvature counterexample

Let  $n \geq 2$ ,  $0 < c < 1$ , and use  $D_j = -i\partial_{x_j}$  on the flat analytic torus  $\mathbb{T}^n$ . Define

$$P = (-\Delta)^{1/2} + cD_1, \quad q(\xi) = |\xi| + c\xi_1. \tag{1}$$

**Theorem 1** (Counterexample independent of convexity terminology). *The operator (1) satisfies every hypothesis printed before Winterrose’s Conjecture 4.1. Its unit principal co-ball is an ellipsoid and has everywhere positive-definite second fundamental form; equivalently,  $D^2(q^2/2)$  is positive definite off the zero section. Nevertheless, for every  $\varepsilon > 0$ , the kernel of  $e^{-\varepsilon P}$  does not extend holomorphically to the tube  $M_\varepsilon^\Phi$  prescribed by conclusion (1). Thus conclusion (4), and hence the full conjecture, is false even under the strong-curvature interpretation of “strictly convex.”*

CONVERGENCE DANS LE DOMAINE COMPLEXE  
DES SERIES DE FONCTIONS PROPRES

par L. BOUTET DE MONVEL

Soit  $X$  une variété analytique réelle compacte de dimension  $n$ ,  $\tilde{X}$  un complexifié de  $X$ ,  $P = P(x, D)$  un opérateur pseudo-différentiel analytique d'ordre 1, elliptique, de symbole positif, sur  $X$  - par exemple  $P = \sqrt{-\Delta}$  où  $\Delta$  est le Laplacien d'une métrique riemannienne à coefficients analytiques. On suppose que  $P$  satisfait à la condition suivante :

- (C) Pour tout  $x \in X$ , l'ensemble des covecteurs  $\xi$  en  $x$  tels que  $\sigma(P)(\xi) \leq 1$  est strictement convexe.

Notons  $\Xi$  le champ hamiltonien du symbole de  $P$ , et  $\Phi = \Phi(t, x, \xi)$  son flot intégral :  $\Phi(t, x, \xi)$  est la valeur à l'instant  $t$  de la solution de l'équation différentielle  $\frac{d}{dt}(x, \xi) = \Xi$  qui vaut  $(x, \xi)$  à l'instant  $t = 0$  (si  $P = \sqrt{-\Delta}$ ,  $\Phi$  est le flot géodésique). Comme le symbole de  $P$  est analytique, et homogène en  $\xi$ ,  $\Phi(t, x, \xi)$  est encore bien défini pour  $(x, \xi) \in T^*\tilde{X}$  assez voisin de  $T^*X$ , et  $t \in \mathbb{C}$  assez petit. Pour  $\varepsilon > 0$  assez petit, soit  $\Omega_\varepsilon \subset \tilde{X}$  l'ensemble des points de base des covecteurs  $\Phi(it, x, \xi)$  avec  $x \in X$  (réel),  $\xi$  réel,  $|\xi| = 1$ ,  $0 \leq t < \varepsilon$ . Un argument facile de perturbation (à partir du cas où  $P$  est un opérateur de convolution sur  $\mathbb{R}^n$ ) montre que, sous l'hypothèse (C),  $\Omega_\varepsilon$  est un voisinage tubulaire strictement pseudo-convexe de  $X$ , de bord  $\partial\Omega_\varepsilon \subset \tilde{C}^\infty$ .

On sait que l'opérateur  $e^{itP}$  est un opérateur intégral de Fourier associé à la relation canonique graphe de  $\Phi(t, \dots)$  ; comme  $P$  est analytique, le noyau  $K(t, x, y)$  de cet opérateur est encore bien défini pour  $x, y$  complexes voisins de  $X$ , et  $t$  complexe assez petit, avec une singularité sur la projection dans  $\tilde{X} \times \tilde{X}$  du graphe de  $\Phi$ . Faisons en particulier

Figure 2: Boutet de Monvel's set-theoretic condition (C), followed by the asserted smooth strictly pseudoconvex tube conclusion.

*Proof.* On the Fourier basis  $e_k(x) = e^{ik \cdot x}$ ,

$$Pe_k = q(k)e_k = (|k| + ck_1)e_k.$$

Hence  $P$  is an analytic classical polyhomogeneous pseudodifferential operator of order  $d = 1$ , formally self-adjoint and nonnegative. Its principal symbol is positive and elliptic because

$$(1 - c)|\xi| \leq q(\xi) \leq (1 + c)|\xi|.$$

It has no negative eigenvalues. Its symbol is independent of  $x$ , so its real Hamilton flow is complete and entire in time.

The unit co-ball is much stronger than segment-strict. Squaring  $|\xi| \leq 1 - c\xi_1$  gives

$$(1 - c^2) \left( \xi_1 + \frac{c}{1 - c^2} \right)^2 + |\xi'|^2 \leq \frac{1}{1 - c^2}. \quad (2)$$

Conversely every point of this ellipsoid has  $1 - c\xi_1 > 0$ , so (2) is exactly  $\{q \leq 1\}$ . It is a smooth ellipsoid with positive curvature everywhere.

For completeness, put  $r = |\xi|$  and  $u = \xi/r$ . For any  $\eta \neq 0$ ,

$$D^2 \left( \frac{q^2}{2} \right) (\xi)[\eta, \eta] = (u \cdot \eta + c\eta_1)^2 + (1 + cu_1)(|\eta|^2 - (u \cdot \eta)^2) > 0.$$

Indeed the second summand is positive unless  $\eta$  is parallel to  $u$ , and in the parallel case the first is positive because  $1 + cu_1 \geq 1 - c > 0$ .

Let  $F = q^2/2$ . The Hamilton flow of  $q$  is  $\varphi_t(x, \xi) = (x + t\nabla q(\xi), \xi)$ , and therefore the map prescribed in the conjecture is

$$\Phi(x, \xi) = x + iq(\xi)\nabla q(\xi) = x + i\nabla F(\xi). \quad (3)$$

Positive definiteness just proved makes it a real-analytic Legendre diffeomorphism. In particular, along the positive first axis,

$$\nabla F(te_1) = (1 + c)^2 te_1, \quad q(te_1) = (1 + c)t.$$

Taking  $t = \varepsilon/(1 + c)^2$  shows that

$$z = x + i\varepsilon e_1 \in M_\varepsilon^\Phi; \quad (4)$$

the inequality  $q(te_1) = \varepsilon/(1 + c) < \varepsilon$  is strict.

The Schwartz kernel on the real torus is

$$K_\varepsilon(x, y) = (2\pi)^{-n} \sum_{k \in \mathbb{Z}^n} e^{-\varepsilon q(k)} e^{ik \cdot (x - y)}. \quad (5)$$

Suppose it extended holomorphically to the point (4) for a fixed  $y$ . Restricting the extension to the real  $x$ -torus at height  $v = \varepsilon e_1$ , uniqueness of analytic continuation forces its  $k$ th Fourier coefficient to be

$$e^{-\varepsilon q(k)} e^{-k \cdot v} e^{-ik \cdot y}.$$

For  $k = -me_1$ ,  $m \rightarrow \infty$ , its modulus is

$$e^{-\varepsilon m(1 - c)} e^{\varepsilon m} = e^{c\varepsilon m}.$$

These coefficients grow exponentially. They cannot be the Fourier coefficients even of a distribution, much less of a real-analytic function. This contradiction proves failure of conclusion (4).  $\square$

*Remark 1* (Scope of the new obstruction). Boutet de Monvel’s original note is formulated for scalar differential operators. Positivity forces their order to be even, and their homogeneous principal polynomial is then even in  $\xi$ . Thus reversibility is automatic in that original scope. Winterrose explicitly changes the statement to general  $P \in \Psi_{\text{phg}}^d(M)$ , where formal self-adjointness only makes the principal symbol real, not even. Theorem 1 refutes that published pseudodifferential formulation. If “the intended theorem” also silently retains the original differential-operator scope, then the example instead identifies a second missing hypothesis in the generalization.

## 4 The segment-strict curvature counterexample

The earlier ambiguity is real and yields a separate obstruction. On  $\mathbb{T}^2$  take

$$P = \partial_1^4 + \partial_2^4, \quad p(\xi) = \xi_1^4 + \xi_2^4, \quad q = p^{1/4}. \quad (6)$$

This is a nonnegative, formally self-adjoint analytic elliptic differential operator. Its  $\ell^4$  unit co-ball has no boundary segments because  $t \mapsto |t|^4$  is strictly convex.

**Proposition 1** (Failure under the literal convex-set convention). *For (6), the flow map in conclusion (1) is not a local diffeomorphism at any nonzero axial covector, for any radius. Its image boundary is only  $C^1$  at the corresponding axes, so conclusion (2) also fails.*

*Proof.* Here

$$F(\xi) = \frac{1}{2}q(\xi)^2 = \frac{1}{2}(\xi_1^4 + \xi_2^4)^{1/2}, \quad \Phi(x, \xi) = x + i\nabla F(\xi).$$

At  $(a, 0)$ ,  $a \neq 0$ ,

$$D^2F(a, 0) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad D\Phi = \text{diag}(I_2, D^2F),$$

so  $D\Phi$  has rank three. Shrinking the tube cannot remove this defect.

The image is the dual-gauge tube with

$$q^*(v) = (|v_1|^{4/3} + |v_2|^{4/3})^{3/4}.$$

Near  $(\varepsilon, 0)$  its boundary is

$$v_1 = (\varepsilon^{4/3} - |v_2|^{4/3})^{3/4} = \varepsilon - \frac{3}{4}\varepsilon^{-1/3}|v_2|^{4/3} + O(|v_2|^{8/3}).$$

The second derivative is unbounded at zero, so the boundary is not  $C^2$  and cannot be real analytic.  $\square$

## 5 Complete translation-invariant characterization

The two examples give all the obstructions in the flat constant-coefficient class, including classical lower-order terms. Let  $q : \mathbb{R}^n \rightarrow [0, \infty)$  be positive, one-homogeneous and real analytic away from zero, with strictly convex unit co-ball. It need not be even. Let  $\mu$  be a positive real-analytic classical symbol of order one with principal part  $q$ , so

$$\mu(\xi) = q(\xi) + r(\xi), \quad r \in S_{\text{cl}}^0. \quad (7)$$

Changing finitely many lattice values is harmless. For  $d > 0$ , define

$$Pe_k = \mu(k)^d e_k. \quad (8)$$

Then  $P$  is a nonnegative, self-adjoint, translation-invariant classical operator of order  $d$ , with principal symbol  $p = q^d$  and  $P^{1/d}e_k = \mu(k)e_k$ .

Write  $F = q^2/2$  and let

$$q^*(v) = \sup_{q(\xi)=1} v \cdot \xi$$

be the polar gauge. Reversibility means  $q(-\xi) = q(\xi)$ .

**Theorem 2** (Exact flat-torus repair). *For the class (8), the following are equivalent:*

- (i) *all six conclusions of Winterrose's Conjecture 4.1, with its printed positive imaginary-time map, hold for some (equivalently every)  $\varepsilon > 0$ ;*
- (ii)  *$D^2F(\xi)$  is positive definite for every  $\xi \neq 0$  and  $q$  is reversible.*

*If only  $D^2F > 0$  is assumed, the kernel is holomorphic on the reflected tube*

$$\mathbb{T}^n + i\{v : q^*(-v) < \varepsilon\},$$

*and all six conclusions hold after replacing the positive imaginary-time map by the negative one,  $\Phi_-(x, \xi) = x - i\nabla F(\xi)$ .*

*Proof.* For every  $q$  in the class, the positive imaginary-time flow is

$$\Phi_+(x, \xi) = x + i\nabla F(\xi), \quad D\Phi_+ = \text{diag}(I_n, D^2F(\xi)). \quad (9)$$

Since the unit co-ball is convex,  $F$  is convex and  $D^2F$  is positive semidefinite. Conclusion (1) therefore forces it to be positive definite.

Assume this positivity for the moment. Fenchel duality gives

$$F^*(v) = \frac{1}{2}q^*(v)^2, \quad \nabla F : \mathbb{R}^n \setminus 0 \longrightarrow \mathbb{R}^n \setminus 0$$

as a real-analytic diffeomorphism with inverse  $\nabla F^*$ , and  $q^*(\nabla F(\xi)) = q(\xi)$ . Hence the tube in conclusion (1) is

$$M_\varepsilon^{\Phi_+} = \mathbb{T}^n + iD_\varepsilon^+, \quad D_\varepsilon^+ = \{v : q^*(v) < \varepsilon\}. \quad (10)$$

The kernel is

$$K_\varepsilon(z, y) = (2\pi)^{-n} \sum_{k \in \mathbb{Z}^n} e^{-\varepsilon\mu(k)} e^{ik \cdot (z-y)}. \quad (11)$$

Because  $\mu(k) = q(k) + O(1)$ , its maximal open tube of normal Fourier convergence is

$$D_\varepsilon^- = \{v : q^*(-v) < \varepsilon\} = -D_\varepsilon^+. \quad (12)$$

More explicitly, if  $q^*(-v) > \varepsilon$ , lattice directions approximate a support direction for  $-v$ , and the corresponding coefficients in (11) grow exponentially. They are not Fourier coefficients of a distribution. Thus conclusion (4) on all of  $D_\varepsilon^+$  implies

$$q^*(-v) \leq q^*(v) \quad (v \in \mathbb{R}^n).$$

Applying this to  $-v$  gives equality. Hence  $q^*$ , and therefore  $q$ , is even. This proves necessity of both conditions.

Conversely, assume  $D^2F > 0$  and reversibility. Then  $D_\varepsilon^+ = D_\varepsilon^-$ , so (9) and Legendre duality prove conclusion (1). A real-analytic defining function near the boundary is

$$\rho(x + iv) = 2F^*(v) - \varepsilon^2.$$

Its complex Levi form is a positive multiple of  $D^2F^*(v)$ , proving the real-analytic boundary and strict pseudoconvexity in conclusions (2)–(3). Duality gives normal convergence of (11), with all derivatives, on compact subsets of the tube, proving conclusion (4).

On  $q^*(v) = \varepsilon$ , write

$$e^{-\varepsilon\mu(\xi)} = e^{-\varepsilon q(\xi)} b_\varepsilon(\xi), \quad b_\varepsilon = e^{-\varepsilon(\mu-q)} \in S_{\text{cl}}^0.$$

The amplitude  $b_\varepsilon$  is elliptic. Poisson summation gives, modulo a smooth kernel, the positive complex phase

$$\phi(x, v, y, \xi) = (x - y) \cdot \xi + i(\varepsilon q(\xi) + v \cdot \xi).$$

Its imaginary part is nonnegative and vanishes exactly at  $x = y$  and  $v = -\varepsilon \nabla q(\xi)$ . Positive  $D^2F$  makes the canonical graph nondegenerate. With  $n$  phase variables, target dimension  $2n - 1$ , source dimension  $n$ , and order-zero amplitude, the standard order formula gives

$$\frac{n}{2} - \frac{(2n - 1) + n}{4} = -\frac{n - 1}{4}.$$

This proves conclusion (5) by the positive complex-phase calculus [4, 5].

Finally, for any smooth positive boundary density, the  $k$ th eigenvalue of  $S_\varepsilon^* S_\varepsilon$  is

$$e^{-2\varepsilon\mu(k)} \int_{q^*(v)=\varepsilon} e^{-2k \cdot v} a(v) d\sigma(v).$$

Laplace's method at the unique nondegenerate support point  $v = -\varepsilon \nabla q(k)$  makes this an elliptic classical symbol of order  $-(n - 1)/2$ ; the order-zero factor from  $\mu - q$  does not change the order. Thus

$$\|S_\varepsilon f\|_{H^{s+(n-1)/4}} \asymp \|f\|_{H^s}.$$

The range is closed and contains every holomorphic Fourier polynomial, since all Fourier multipliers are nonzero. Such polynomials are dense in the source paper's definition of  $\mathcal{O}^{s+(n-1)/4}(\partial M_\varepsilon^{\Phi+})$ , so the range is the entire space. This is conclusion (6).

If reversibility is dropped, repeat the same argument on  $D_\varepsilon^- = -D_\varepsilon^+$  with  $\Phi_-(x, \xi) = x - i\nabla F(\xi)$ . The phase and its critical support point are then aligned exactly as above, proving the sign-corrected statement.  $\square$

## 6 What this says about the intended theorem

There are now three distinct readings, and the result is useful in each.

1. Under the literal no-segments definition, the  $\ell^4$  differential operator directly refutes conclusions (1)–(2).
2. If “strictly convex” means positive curvature, the  $\ell^4$  example is excluded, but the Randers multiplier still refutes Winterrose's printed pseudodifferential formulation through conclusion (4).



3. If the intended statement silently combines positive curvature with Boutet de Monvel’s original differential-operator scope, both obstructions have been excluded by intent. In that case this packet is a precise statement correction rather than a refutation of the intended theorem. Theorem 2 remains a complete result: it proves all six conclusions for every strongly convex reversible classical multiplier, including arbitrary classical lower-order terms, and identifies the exact additional repair needed for nonreversible pseudodifferential symbols.

The variable-coefficient, strongly convex, reversible case on a general analytic manifold remains open. The geometric part should be locally stable under small radius because the leading Legendre Hessian is nondegenerate, but conclusions (4)–(6) require the global analytic half-wave/FIO construction that Winterrose’s thesis itself identifies as the unresolved core. No claim to solve that general problem is made here.

## 7 Verification and novelty notes

1. **Hypotheses and failures.** Every printed condition was checked for the Randers operator, including strong convexity and the flow assumptions; its failure is exponential Fourier growth. The  $\ell^4$  failure is independently a rank defect and a non- $C^2$  boundary.
2. **Version audit.** ArXiv v1–v3, the published text, the original note, and thesis Sections 7.1–7.2 contain neither a convexity definition nor an evenness hypothesis.
3. **Bounded novelty search.** On 22 July 2026 the run indexes and web/arXiv were searched by the source id/title and combinations of strict/strong convexity,  $\ell^4$ , Randers, nonreversibility,  $|\xi| + c\xi_1$ , and imaginary Hamilton flow. No prior counterexample or this characterization was found; this is not exhaustive certification.

**Human-review recommendation.** Prioritize Theorem 1: check the source’s flow sign and its standard inclusion of the analytic classical multiplier  $|D| + cD_1$ . Then review the complex-phase and Laplace steps in Theorem 2. Author clarification can still decide whether the intended scope retained both strong curvature and differential operators.

## References

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